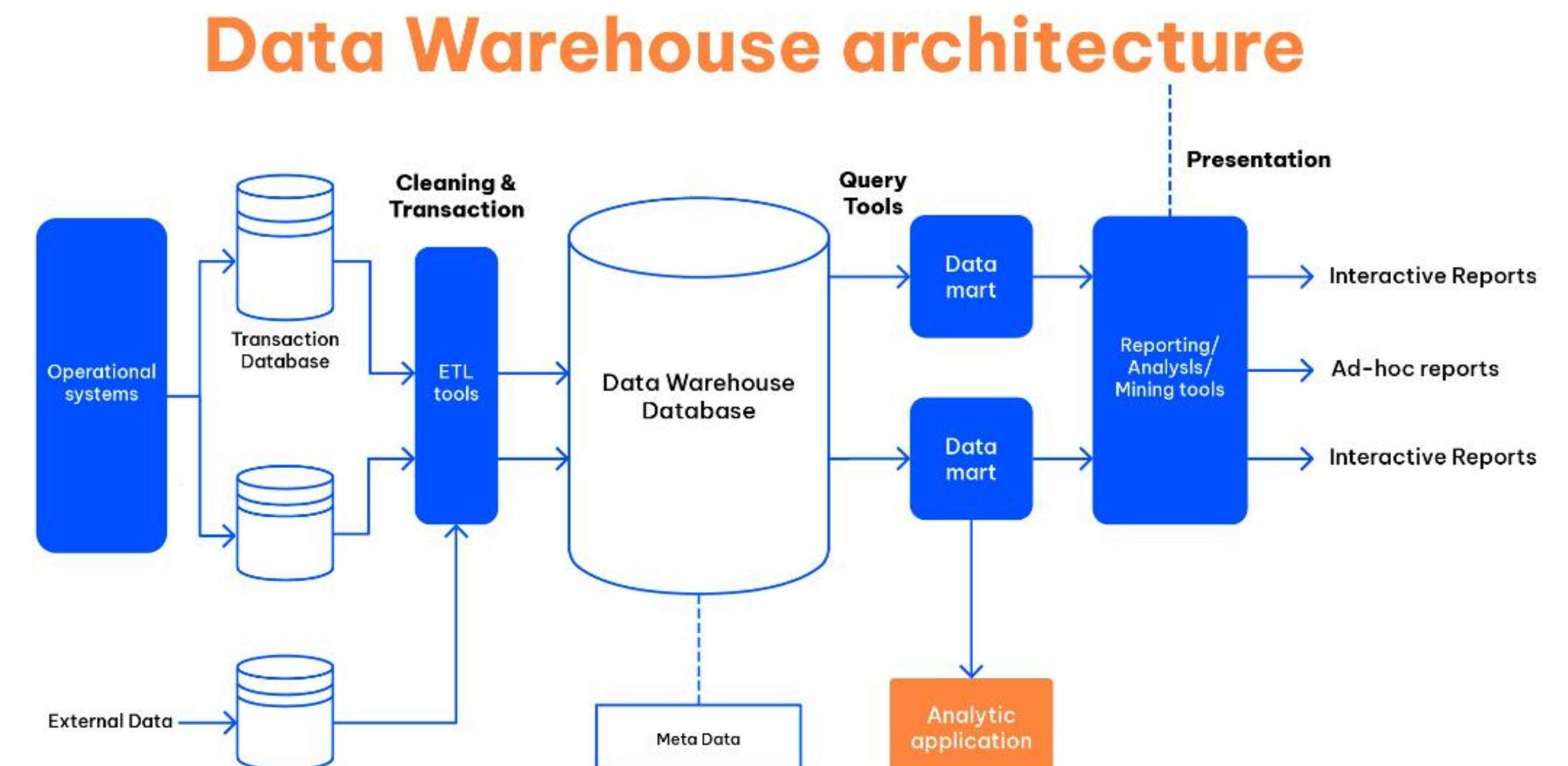


| ARCHITECTURAL SHIFT: ETL TO ELT

 **The Traditional ETL:** Data is transformed on a middle server **before** loading into the warehouse. Often creates a bottleneck and adds infrastructure costs.

 **The Modern ELT:** Data is loaded raw into the cloud warehouse first. Transformation happens **inside** the warehouse (Snowflake, BigQuery, etc.).

 **The Matillion Edge:** Matillion uses a **Push-Down Architecture**, generating optimized SQL to leverage the native compute power of your cloud platform.



| STRATEGIC CHOICE: WHEN TO USE WHICH?

Factor	Legacy ETL	Modern ELT (Matillion)
Data Volume	Small to Mid-sized datasets.	Huge volumes & Big Data.
Scalability	Fixed by hardware servers.	Elastic (Grows with the cloud).
Speed	Sequential; can be slow.	High-speed parallel processing.
Compliance	Masking *before* storage.	Agile transformations after load.
Complexity	Rigid and code-heavy.	Agile, GUI-driven & Scalable.

Verdict: Choose **ELT with Matillion** for cloud-first strategies where performance, schema agility, and cost-efficiency are prioritized over legacy rigid pipelines.

| IMAGE SOURCES



Thumbnail
for gem-
corp.tech

<https://gem-corp.tech/wp-content/uploads/2024/08/data-warehouse-architecture-webp.webp>

Source: gem-corp.tech